

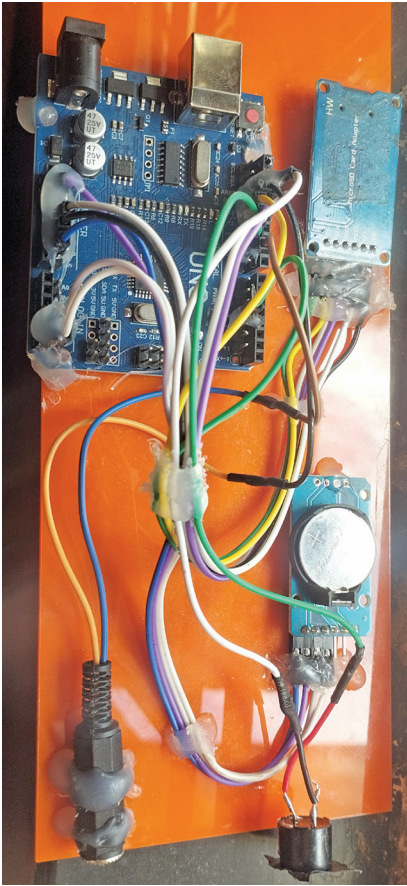


Fig. 4: Author's prototype

the SD card. It carries data from the microcontroller to the SD card when writing data

5. SCK (serial clock): This pin is the clock signal generated by the microcontroller for synchronising data transfer between the microcontroller and the SD card

6. CS (chip select): This pin is used to enable communication with the SD card. When this pin is set to low (usually connected to ground), it indicates to the SD card that it should listen to commands from the microcontroller. The playAudio function is called to play the audio file stored on the SD card when a bell ringing event occurs

#### Real-time clock (RTC) module.

The real-time clock (RTC) module typically consists of several pins, each serving a specific function:

**TABLE 2**  
**CONNECTING COMPONENTS MicroSD**

| Arduino Pins   | Pins MicroSD Module and Buzzer                |
|----------------|-----------------------------------------------|
| Arduino pin 4  | CS (chip select) pin-6                        |
| Arduino pin 9  | Positive pin of audio female connector        |
| Arduino pin 10 | Connected to positive pin of buzzer           |
| Arduino pin 11 | Connected to MOSI pin-4 (master out slave in) |
| Arduino pin 12 | Connected to MISO pin-3 (master in slave out) |
| Arduino pin 13 | SCK-pin-5 (serial clock)                      |

**TABLE 3**  
**RTC MODULE CONNECTION**

| RTC Module | Arduino Pins                         |
|------------|--------------------------------------|
| SDA        | Connected to Arduino analogue pin-A4 |
| SCL        | Connected to Arduino analogue pin-A5 |
| Vcc        | Connected to Arduino analogue pin-A5 |
| GND        | Connected to Arduino GND pin         |

1. VCC: This pin is connected to the power supply (usually 5V or 3.3V) to provide power to the RTC module

2. GND: This pin is connected to the ground (0V) of the power supply to complete the circuit

3. SDA (serial data): This pin is used for bidirectional serial data transfer between the RTC module and the microcontroller. It carries data between the microcontroller and the RTC module during communication

4. SCL (serial clock): This pin is used to synchronise the data transfer between the microcontroller and the RTC module. It generates clock pulses for timing the serial communication

5. SQW (square wave output) not connected: This pin outputs a square wave signal with a frequency specified by the RTC module. It can be used as an interrupt signal or for other timing-related purposes

6. 32kHz output not connected:

Some RTC modules may have an additional pin for outputting a 32kHz clock signal, which can be used for timekeeping or other timing-sensitive applications. These pins allow the RTC module to communicate with and be controlled by the microcontroller, enabling

accurate timekeeping and other time-related functions in embedded systems

## Construction and testing

For the construction of the device, first upload the source code `code.ino` into the Arduino Uno board. Then assemble all the components according to the circuit diagram as shown in Fig. 3 and Tables 2 and 3. The author's prototype is shown in Fig. 4.

After proper assembly of the device, the call bell is ready to work as follows:

Initially, the circuit is powered up with a 12V external power supply. When school starting time arrives, an announcement for the prayer or national anthem will play. After that, the bell will ring immediately to indicate the start of school. At the end of the first period (as per the set time in the program), the bell will ring, indicating the end of the first period and the start of the second period.

Similarly, the designed circuit will track the end of each period and ring the bell accordingly. Repeat from step 1.

As an improvement, the bell sound can be replaced with an audio announcement after each period, where the period number is mentioned. **EFY**

### EFY Note

The source code of this project is available

for free download at [www.electronicsforu.com](http://www.electronicsforu.com)



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